Lead exposure and IQ loss in children: a Bayesian and frequentist meta-analysis of recent epidemiological studies

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Table of contents

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Notes

- Journal: Population Health Metrics, BMC
- Using PRISMA 2020 checklist for structure
- Using raRoB tool for Risk of Bias Assessment

Abstract

Background

Lead (Pb) is a pervasive environmental pollutant originating from anthropogenic and natural sources. Although concentrations have declined globally in past decades following the phase-out of leaded gasoline, lead remains present in the environment and human blood (1,2). In Europe, primary exposure sources include dietary products, drinking water, and indoor dust (3,4). The most recent Global Burden of Disease study estimated that in 2023, 3.4 million (95% Uncertainty Interval (UI): 2.7; 4.0 million) Disability-Adjusted Life Years (DALYs) were attributable to lead exposure in the European Union (EU) (5). This highlights the continued public health significance even of low-level exposure.

Exposure to lead is associated with multiple adverse health outcomes across the human life course, including decreased kidney function (6), liver disease (7), coronary artery atherosclerosis (8), cardiovascular mortality (9), infertility in women (10), attention-deficit/hyperactivity disorder in children (11), and antisocial behavior (12). Among these endpoints, impaired cognitive development in children — manifested as reduced IQ scores — (13–16) represents a particularly critical endpoint for public health policy. Children are especially vulnerable to lead's neurotoxic effects due to their developing nervous systems and relatively high exposure per body mass. Because lead-associated IQ decrements are irreversible and persist throughout life, they lead to substantial and enduring societal costs (17), making continued efforts to accurately characterize the exposure-response relationship important for evidence-based policy.

Contemporary blood lead levels (BLL) in the EU have declined markedly and now predominantly fall below 5 µg/dL (HBM4EU Dashboard), well below the concentrations examined in many earlier studies (<10 µg/dL). Importantly, there is no evidence for a threshold concentration below which the effects of lead on the IQ can be assumed to be negligible (18,19). There is evidence for a supralinear exposure-response relationship with steeper IQ decrements per unit increase in BLL at low concentrations compared to higher concentration ranges (16,20). These findings highlight the need for an updated quantitative synthesis that reflects both recent epidemiological evidence and the lower exposure range now predominant in European populations, incorporating recent data to inform risk assessment at currently relevant exposure levels.

Existing burden of disease assessments at national and global scales (5,17,21–23) have predominantly relied on the foundational study by Lanphear et al. (2005), a pooled analysis of seven cohort-studies, or subsequent re-analyses of the same data (16,24). While this landmark studies has substantially shaped research and policy on childhood lead exposure, methodological issues and errors have been raised (24,25). These limitations, combined with the availability of more recent epidemiological data, underscore the need not only for an updated evidence synthesis but also for methodologically robust approaches that can provide reliable exposure-response estimates for burden of disease and risk assessments.

Methodological advances in meta-analysis offer an opportunity to enhance the utility of synthesized estimates for burden of disease and risk assessment applications. While frequentist meta-analysis has been a conventional approach in environmental epidemiology, Bayesian meta-analysis provides a coherent framework for uncertainty quantification through posterior distributions that inherently incorporate both within-study and between-study variability (26). Unlike frequentist confidence intervals, which are often misinterpreted as probabilistic (27), Bayesian credible intervals directly represent the probability distribution of the parameter of interest, facilitating transparent communication of uncertainty. Furthermore, Bayesian posterior distributions can be directly propagated into probabilistic burden of disease and risk assessments, providing seamless integration with probabilistic frameworks in public health decision-making. By comparing frequentist and Bayesian approaches, we will explore how inferential framework and uncertainty quantification influence exposure-response estimates and their application.

The aim of this study is twofold: first, to synthesize and quantitatively pool available epidemiological findings on the association between blood lead levels and IQ scores in children; and second, to conduct both a frequentist and Bayesian meta-analyses, comparing how these analytical frameworks influence exposure-response estimates, uncertainty characterization, and applicability to burden of disease and risk assessment.

Methods

- ☒ Overview: systematic review and meta-analysis, comparing a frequentist to a Bayesian approach
- ☒ Inclusion & exclusion criteria, databases, search strategy, screening process
- ☒ Risk of bias analysis method
- ☒ Description & rationale of harmonization of reported effect measures
- ☒ Description of meta-analysis methodology (frequentist & Bayesian), models and software used, heterogeneity identification, prior choice
- ☒ Sensitivity analysis

Literature Review

This review adheres to the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines (28). The checklist can be found in the supplementary file 1. The reporting of the Bayesian methods and their results was guided by the Bayesian Analysis Reporting Guidelines (BARG) (29).

We conducted a literature review to identify published studies investigating the association between postnatal (concurrent) exposure to Pb and IQ loss in children. This study was conducted within the EU-funded project *Partnership for the Assessment of Risks from Chemicals (PARC)*. The literature review presented here was used in multiple studies within PARC, including one

Table 1: Search queries in Pubmed and Scopus

...1	Concepts				Detailed queries
Pubmed	Substance incl. Mesh	IQ, neurodev, disability	Association, ERF	Population	((Pb[Title/Abstract]) OR (lead[MeSH Terms]))
Pubmed	Substance + title only	IQ, neurodev, disability	Association, ERF	Population	((Pb[Title/Abstract]) OR (lead[Title/Abstract])
Scopus	Substance incl. Mesh	IQ, neurodev, disability	Association, ERF	Population	(TITLE-ABS("Pb") OR CHEMNAME("lead"))
Scopus	Substance + title only	IQ, neurodev, disability	Association, ERF	Population	(TITLE-ABS("Pb") OR TITLE-ABS("lead")) A

on the association of (methyl)mercury and lead on IQ loss for multiple exposure timeframes. Here, we will only report results for concurrent lead exposure and IQ loss, but the figures and tables documenting the literature review do contain parts of the search for other associations and exposure timeframes.

We searched for articles published between January 2005 and July 2023. The year 2005 was chosen because, as described in the background section, burden of disease studies predominantely used a study published in 2005 (16) and we aimed at updating the evidence base with recently published studies. Also, lead concentrations have decreased significantly in the past decades and we aimed at identifying epidemiological studies conducted in populations with currently relevant exposure levels.

Search queries in Pubmed and Scopus included the following terms: substance (Pb - including Mesh terms), outcome (IQ and neurodevelopment), population (children) and results (association, ERF) (Table 1).

All titles and abstracts were screened by two reviewers. Any conflicts were resolved by a third person. Full text was then assessed by one reviewer. Publications were considered relevant according to predefined PECOTS – Population (children aged 0 to 11), Exposure (biological measures of Pb from all matrices), Comparator (no exposure or lower levels of exposure), Outcome (IQ), Timing of exposure (no restriction) and Setting (no restriction) (Table 2). We excluded reviews and background articles that inventoried the literature with no new results. We also excluded references written in languages other than English.

As described above, the literature review was conducted with as the basis for multiple studies and thus had a larger scope. For this study, only studies that used a continuous, log-transformed BLL scale were included, so that effect estimates could be harmonized. The exposure timeframe for this study was restricted to concurrent, meaning that IQ and BLL measurement must be conducted at the same time. Descriptive study data was extracted using a grid based on the one proposed by NTP OHAT (30). For all included studies, effect estimates from the most-adjusted model were extracted and the set of confounders adjused for recorded.

Risk of Bias Assessment

To assess study quality, we conducted a Risk of Bias assessment using the rapid Risk of Bias (RoB) tool with a specific focus on human observational epidemiological studies (31). Each included publication was assessed for RoB in the domains selection, exposure, outcome, confounding, follow-up, analysis, and selective reporting. In total, 15 items were evaluated for each study. The overall RoB for each individual study was then categorized by one researcher as either low, moderate or high, following two different summary metrics provided by the RoB tool (for more information, see (31)). Each publication was first assessed by one researcher. Subsequently, consensus with a large language model (LLM) was sought. This was done by uploading the list of all raRoB items along with the respective publication. Then, the LLM was promted to assess the study, using the raRoB items and possible answer categories. If there were any deviations in the judgement of the researcher and the LLM, the respective item was revisited by the researcher and they made the final decision.

Harmonization of Effect Estimates

Studies investigating the association between lead exposure and IQ in children have made different assumptions regarding the shape of the exposure-response relationship. While some studies assume a linear relationship (32,33), this assumption may only be valid at very low concentrations (24). Studies that tested multiple shapes of the exposure-response function have concluded that a log-linear relationship, where BLL are logarithmically transformed, provides a more appropriate model (24,34).

The general form of regression model used for a log-linear association is:

$$y = \alpha + \beta \cdot log_b(X) + \varepsilon \tag{1}$$

Where y represents the IQ score, α is the intercept, β is the regression coefficient (slope), X denotes BLL, b is the logarithmic base, and ε is the error term.

In the literature, different logarithmic bases are used to transform BLL: the natural logarithm ($\log_e$ where $e \approx 2.71828$), base 2, and base 10. To enable pooling of the effect estimates, we harmonized all regression coefficients and their standard error (SE) to the natural logarithm scale ($\log_e$). The calculations are documented in suppl. file 2 (Screening documentation excel) In some publications, the BLL scale was left untransformed, so on a linear scale. These were excluded from the meta-analysis, as transformation of effect estimates for linear BLL to log-linear could not be justified based on visual inspection (supplement: plots?).

Meta-Analysis Methodology

In order to quantitatively synthesize the available epidemiological literature on the association of children's blood lead levels and IQ, we conducted a meta-analysis of the results, comparing a Bayesian to a frequentist approach.
In both approaches, the same effect estimates from the identified publications were included and analogous models were chosen to enable comparison between the results. We estimated pooled effect estimates for the loss of IQ points per continuous unit increase in $\log_e$-transformed blood lead concentration (µg/dL).

R version 4.4.2 (2024-10-31) (35) was used for all computations. All computational code for this study is publicly available on GitHub.

Bayesian Meta-Analysis

For the Bayesian approach, we employed a random-effects meta-analysis model with two hierarchical levels. At the first level, the observed effect size from each study $\hat{\theta}_k$ is modeled as an estimate of that study's true underlying effect θ_k, with deviation from the true effect determined by the within-study sampling error σ_k (treated as known based on reported study statistics). At the second level, these study-specific true effects are themselves assumed to vary across studies following a normal distribution, characterized by an overall pooled effect μ and between-study standard deviation (heterogeneity) τ.

$$\begin{aligned} \hat{\theta}_k &\sim \mathcal{N}(\theta_k, \sigma_k) \\ \theta_k &\sim \mathcal{N}(\mu, \tau) \end{aligned} \tag{2}$$

Where:
$\hat{\theta}_k$ = Observed effect size for study k
σ_k = Within-study sampling standard deviation for study k
θ_k = True underlying effect for study k
μ = Overall pooled effect
τ = Between-study standard deviation (heterogeneity)

The Bayesian model was fit using the R package `brms` (36) version 2.22.0. Posterior draws were sampled using Markov chain Monte Carlo (MCMC) and the No-U-Turn sampler (NUTS) (37). We ran four chains with 4,000 iterations each, of which 2,000 iterations were discarded as warmup, resulting in 8,000 total post-warmup draws. MCMC convergence was assessed using the Gelman-Rubin statistic ($\hat{R}$) and chain resolution was assessed using the effective sample size (ESS). As shown in Equation 2, heterogeneity τ was modelled as a parameter directly and I^2 was calculated using the typical within-study variance that is used in the frequentist approach so that it is comparable.

Priors

In Bayesian statistics, prior knowledge or assumptions from previously published studies, domain knowledge or beliefs are updated by the newly observed data and together result in a posterior distribution. Prior distributions must be chosen for all parameters to be estimated in the analysis, which in this case are the overall pooled effect estimate μ and between-study standard deviation (heterogeneity) τ. For both parameters, weakly-informative / conservative priors were chosen so that the posterior distributions are softly being bounded to a realistic range. Because the unit scale of the included effect estimates was harmonized to IQ shift per unit increase in $\log_e$-transformed blood lead concentration (µg/dL), the priors will be specified on the same scale.

For the overall pooled effect μ, a normally distributed prior centered at -1 with a standard deviation (SD) of 2 was chosen. This translates to the assumption that the overall pooled effect lies between -5 and 3 with about 95% probability. The center point of the prior was chosen to be -1 because the negative effects of lead on the developing nervous system and its toxicological mechanisms of action are well established in the scientific literature (18).
Since between-study standard deviation (heterogeneity) τ must by definition be positive, a truncated normal distribution centered at 0 with a SD of 2 was specified. This translates to the assumption that τ is small, but due to its wide SD, the prior is still vague and thus weakly-informative. In summary, the following priors were specified for the Bayesian model:

$$\begin{aligned} \mu &\sim \mathcal{N}(-1, 2) \\ \tau &\sim \mathcal{N}^+(0, 2) \end{aligned} \tag{3}$$

Prior and posterior predictive checks were conducted using the `bayesplot` package (38) to assess the appropriateness of the specified priors. Additionally, a sensitivity analysis with alternatice priors was conduced, as described below.

Frequentist Meta-Analysis

For the frequentist meta-analysis we employed a random-effects model, assuming the same multilevel structure as for the Bayesian model Equation 2. Parameters were estimated via the restricted maximum-likelihood (REML) estimation from the R package `metafor` version 4.8-0 (39). The I^2 statistic was calculated as a measure of heterogeneity, representing the proportion of total variation attributable to between-study heterogeneity τ^2.

Sensitivity Analysis

We conducted two sets of sensitivity analyses. To examine the sensitivity of the results to possibly biased study estimates, we ran both the frequentist and Bayesian models again twice, excluding (1) studies with a high RoB and (2) with a high or medium RoB.

To examine the sensitivity of the results to the choice of prior distributions, we fit the Bayesian model again twice, with alternative priors for both the main effect (μ) and heterogeneity (τ). The first set of priors was more informative (more narrow) than the main priors, with $\mu \sim \mathcal{N}(-1, 1)$ and $\tau \sim \mathcal{N}^+(0, 1)$. The second set of priors was less informative (or broader) with $\mu \sim \mathcal{N}(0, 6)$ and $\tau \sim \mathcal{N}^+(0, 6)$. Notice that for the main effect, we centered the broader prior around 0, giving equal weight to the probability of a protective and detrimental effect of lead.

More informative priors (less conservative):

$$\begin{aligned} \mu &\sim \mathcal{N}(-1, 1) \\ \tau &\sim \mathcal{N}^{+}(0, 1) \end{aligned} \tag{4}$$

Less informative priors (more conservative):

$$\begin{aligned} \mu &\sim \mathcal{N}(0, 6) \\ \tau &\sim \mathcal{N}^{+}(0, 6) \end{aligned} \tag{5}$$

Results

- ☒ Cite studies that might appear to meet inclusion criteria, but were excluded & explain why (no citation so far)
- ☒ Study characteristics (table), including original and harmonized effect measures
- ☒ Flow diagram, description of (full text) screening results
- ☒ RoB results for each included study
- ☐ Forest plots
- ☐ Sensitivity analysis results

Literature Review

As described in the methods, the systematic review was conducted with a larger scope, because it was used as a basis for multiple studies, which is why it includes another substance (mercury, Hg). The systematic search in PubMed and Scopus yielded 1,073 records for lead. After duplicate removal, 974 records (Pb & Hg) were screened in abstract and title, which lead to 149 records (Pb & Hg) to be screened in fulltext. After fulltext-screening, 36 records remained for Pb. These were then assessed for their eligibility for the meta-analysis. Twelve studies were excluded because they either transformed the BLL variable into categories or left it on a linear (untransformed) scale. Four studies were excluded because they exclusively assessed the effects of prenatal lead exposure. Furthermore, four studies were excluded because the cohorts included in the study were already included in another study and would have been a duplicate. Four additional studies were excluded because they either solely reported effects of chemical mixtures, because they used non-linear models or failed to report essential statistics. Finally, twelve studies were included in the meta-analysis. Study characteristics and regression coefficients are shown in table 2.

A PRISMA flow diagram summarizing the review process is shown as in figure 1 (28). A file containing detailed information on the screening process and exclusion reasons can be found in suppl. file 2.

Table 2: Study Characteristics

First author, year	Study population (N)	Geographical area	Mean age / age range	BLL indicator	BLL value (µg/dL unless stated otherwise)	IQ measurement instrument
Bauer 2020	635	Province of Bresia, Italy	10-14	Median	1.3	WISC-III (FSIQ)
Crump 2013	1355	USA (Boston, Cleveland, Cincinnati, Rochester), Mexico, Australia (Port Pirie) and Yugoslavia	5-7	Mean	10.0	WISC (FSIQ)
Dantzer 2020	224	USA (Greater Cincinnati, Ohio metropolitan region)	12	Mean	0.6	WISC-IV (FSIQ)
Desrochers-Couture 2018	606	Canada (10 different study sites)	3-4	Mean	0.7	WPPSI-III (FSIQ)
Earl 2016	127	Australia, New South Wales, Port Pirie and Broken Hill	7-8	Mean	3.9	WISC-IV (FSIQ)
Hong 2015	839	South Korea (two urban cities, two industrial cities, and one rural district)	8-11	Mean	1.8	Korean WISC (FSIQ)
Lee 2021	502	South Korea (Seoul, Gyeonggi, and Incheon provinces)	6	Mean	1.4	Korean WISC (FSIQ)
Lucchini 2019	299	Southern Italy (City of Taranto, Puglia region, 5 sub-areas, incremental distance from a highly industrial area)	6-12	Median	8.4 ng/mL	WISC-IV (FSIQ)
Lucchini 2012	299	Northern Italy (Valcamonica and Garda Lake regions)	11-14	Median	1.5	WISC-III (FSIQ)
Menezes-Filho 2018	155	Brazil (Municipality of Simões Filho, metropolitan region of Salvador, Bahia)	7-12	Mean	1.6	WASI (FSIQ)
Roy 2013	708	India (Chennai)	3-7	Mean	11.5	Binet–Kamath Scale of Intelligence (Tamil version of Stanford–Binet) (FSIQ)
Schnaas 2006	150	Mexico (Mexico City)	6-10	Mean	6.7	WISC-Revised (FSIQ)

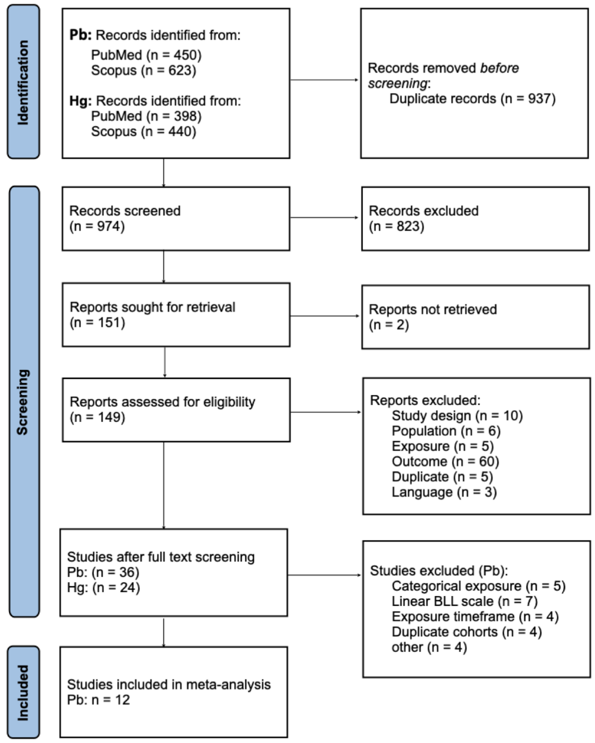


Figure 1: Figure 1: PRISMA flow diagram

Risk of Bias

The rapid RoB assessment resulted in eight studies with a low risk (34), two with a moderate risk (48) and two studies with a high RoB (49).
Domains with a higher RoB were *non-response rate* (Selection), where many of the studies did not report the non-response rate or did not discuss wether non-responding individuals might systematically differ from participants in the studies, leading to selection bias. Another domain with overall higher RoB in the included studies was *accounting for confounding* (Confounding), where important confounders like maternal IQ were left out or the authors used a data-driven selection approach for covariate selection, when a theory-driven approach is preferable (50). Finally, in the domain *handling of missing values* (Analysis), some of the studies gave no information on how they handeled missing values and others handeled them inadequately. Results from the RoB assessment is presented in table 3.

Table 3: Risk of bias assessment results

Study, year	Selection			
	Eligibility criteria	Comparability of exposure and comparison groups	Non-response rate	Recruitm
Bauer 2020	+	+/-	-	
Crump 2013	+	+	+/-	
Dantzer 2020	-	+	-	
Desrochers-Couture 2018	+	+	-	
Earl 2016	+	+/-	-	
Hong 2015	+	+	+/-	
Lee 2021	+	+	+/-	
Lucchini 2019	+	+	-	
Lucchini 2012	+	+	+/-	
Menezes-Filho 2018	+	+	-	
Roy 2013	+	+	-	
Schnaas 2006	+	+	+/-	

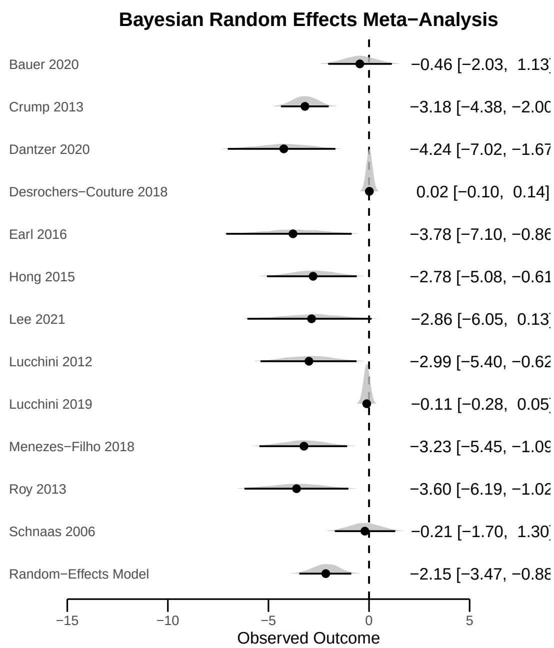
Meta Analysis Results

The prior predictive check indicated that the specified priors are appropriate (suppl. file 3). The posterior predictive check suggested appropriate model specification (suppl. file 4). $\hat{R}$ values indicate well converged MCMC chains (1.00 for both μ and τ). ESS for the central part of the posterior distribution for μ is 1,872 and 3,179 for the tails of the distribution. Both values exceed 1,000, indicating efficiently sampled posterior distribution of μ across both its central tendency and tail regions, and that central point estimates and credible intervals can be interpreted with confidence.

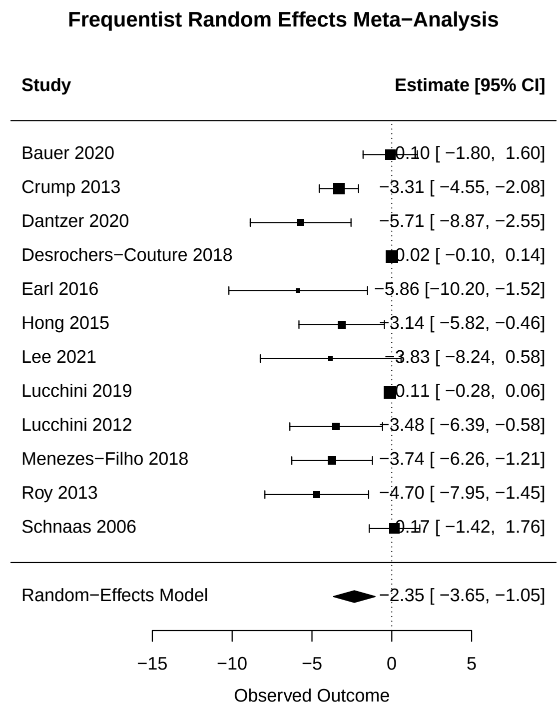
The Bayesian random-effects meta-analysis yielded a posterior mean pooled estimate of -2.15 (95% credible interval (CrI): -3.48; -0.92) IQ points per unit increase in $\log_e$-transformed BLL (µg/dL). The frequentist approach produced a comparable pooled estimate of -2.35 (95% confidence interval (CI): -3.65; -1.05). Substantial between-study heterogeneity was observed under both frameworks. In the Bayesian model, between-study heterogeneity was estimated as $\tau^2 = 4.32$ [95% CrI: 1.28; 10.6], corresponding to an I^2 of 98.3% [95% CrI: 95.8%; 99.5%]. Similarly, the frequentist model estimated $\tau^2 = 3.79$ (95% CI: 1.24; 3.79), with $I^2 = 98.5\%$ (95% CI: 95.65; 99.57%).

Forest plots of both approaches are shown in figure 2:

The study estimates depicted in figure 2 are not the same in both approaches because the Bayesian model estimates a posterior distribution for every individual study that is influenced by all of the included studies as well as the priors. This leads to a shrinkage of the mean coefficients towards the pooled effect and is comparable to the inverse-variance weighting in frequentist meta-analyses.



(a) Figure 2: forest plots



(a) Figure 2: forest plots

Sensitivity Analysis Results

We conducted two sets of sensitivity analyses, the results are shown in table 4. First, in order to test the sensitivity of the results to possibly biased study estimates, we excluded (1) studies with high RoB and (2) with high or medium RoB. For both the frequentist and Bayesian models, the strength of the mean effect decreased after removing studies with high risk of bias and slightly increased again after removing high and medium RoB studies. Because the margin of shift in effects are small and the effects shifted in both directions, we concluded that the results are generally robust to possible individual-study bias.

The second set of sensitivity analyses examined the sensitivity of the results of the Bayesian meta-analysis to prior specification. With more informative (narrower) priors ($\mu \sim \mathcal{N}(-1, 1)$, $\tau \sim \mathcal{N}^+(0, 1)$),the posterior mean μ decreases. This is expected, because the narrow prior restricts the posterior distribution to lower effect ranges.
When using less informative (broader) priors ($\mu \sim \mathcal{N}(0, 6)$, $\tau \sim \mathcal{N}^+(0, 6)$), the posterior mean μ increases slightly. In this case, more "trust" is given to studies reporting high effects with large uncertainty ranges. The results of the prior sensitivity analysis strengthens the priors specified for the main analysis. Suppl. file 5 shows prior, observed data and posterior distributions of all three prior specifications.

Table 1: Table 4: Sensitivity analyses by modeling framework (Bayesian vs Frequentist)

Model type	Sensitivity	Variant	Estimate
Bayesian	Priors	main	-2.17 (-3.50; -0.91)
Bayesian	Priors	narrow	-1.88 (-2.90; -0.89)
Bayesian	Priors	wide	-2.40 (-4.14; -0.91)
Bayesian	RoB	full	-2.19 (-3.51; -0.94)
Frequentist	RoB	full	-2.35 (-3.65; -1.05)
Bayesian	RoB	low	-1.93 (-3.57; -0.36)
Frequentist	RoB	low	-2.23 (-3.91; -0.55)
Bayesian	RoB	low_medium	-1.83 (-3.29; -0.52)
Frequentist	RoB	low_medium	-2.01 (-3.42; -0.61)

Discussion

General interpretation in light of evidence

- ~~updated pooled estimate confirms negative effects of lead exposure on IQ -> compare magnitude of effect to other studies mentioned in background / in literature~~
- ~~heterogeneity in included studies: why? + priors can help to mitigate~~
- ~~are findings generalizeable? Populations in included studies~~

- ~~RoB sensitivity analysis suggested robustness of findings to RoB~~
  - ~~RoB assessment suggests high RoB for study that is used / cited in multiple (EBD) studies~~
- ~~lead - IQ loss good use-case to examine differences and similarities of both approaches~~

In this study, we provide updated pooled effect estimates for the association of children's lead exposure on their IQ. Our results are in line with previous toxicological and epidemiological studies (13,15,16,18) and previous meta-analyses. However, because of methodological differences, the magnitude of the pooled effect cannot be directly compared to previous meta-analyses. For example, in previous studies, the effect of lead has been estimated for increases in defined exposure categories (19,51,52). Also, different coefficients like partial r (14) or standardized differences in mean (53) have been estimated in previous meta-analyses.

We observed substantial heterogeneity in the studies included in this meta-analysis. We identified no single obvious source of heterogeneity. Instead, multiple differences in study population (size, location, age, exposure range) and statistical methods, especially the covariates included in the statistical models (adjustment set) might have caused heterogeneity between the studies (see table 2). Using an IQ test that has not been culturally adapted to the population it is used in, the IQ score results may be an underestimate of the true IQ. This might be the case in (45). Studies that reported a larger standard error of the regression coefficient also tended to reported larger effects. This might indicate publication bias and is one explanation for the substantial heterogeneity in this meta-analysis.
By incorporating prior knowledge and thus laying less "trust" on such studies, Bayesian models are less sensitive to such outliers, especially when pooling a small number of studies.
As mentioned above, the included studies are based on population samples from multiple different regions (North America: 4, South America: 1, Europe: 3, Asia: 3, Australia: 1) and some populations have been sampled in hotspot regions with presumably higher lead concentrations (45). This limits the generalizability of our findings to a specific population.

The RoB sensitivity analysis suggested that the results are generally robust to possible individual-study bias. Importantly, the study published by Crump et al. (2013) (24), which is a re-analysis of the landmark study by Lanphear et al (2005) (16) received a high RoB rating in our assessment. The results of this study have been used in multiple down-stream analyses, including in the most recent Global Burden of Disease study (5). In this study, the burden of disease in 204 countries and for several health risks, including lead exposure was estimated. The main driver for the high RoB identified in the analysis of Crump et al. is the stepwise backward-elimination regression approach, which is central to their findings and has received substantial criticism (50,54,55). This highlights the need for an updated evidence synthesis which we provide in this study.

This study does not only provide an updated effect measure for the association of lead and IQ, but additionally comapres a more conventional, frequentist approach to meta-analysis to a Bayesian approach that is less used in environmental epidemiology to date.

Differences, similarities, strengths & weaknesses of both approaches

- similar pooled estimates (mu and tau / I2)
- shrinkage (vs. inverse-variance weighting) + incorporating prior knowledge causes difference in sensitivity to outliers (good for small number of studies) -> link to beginning of section
- how to think about priors, their effect on the result & how to set them
- freq: many "out of the box" software solutions, requires less programming skills
- Bayes: makes researchers consider choices / assumptions more consciously because explicit
  - like multilevel structure (has to be specified in Bayes approach but is "hidden" in freq approach) or priors (are implicit in freq)

Study limitations & strengths

- limitations:
  - included studies have adjusted for different sets of confounders
- strengths:
  - to our knowledge first comparison of Bayesian & frequentist meta-analysis in field environ epi
  - all data & code available to public

Outlook

- possible use-cases of probabilistic meta-analysis (risk assessment, EBD)
  - probabilistic EBD (give examples of deterministic approach?)
- full posterior distributions for mu and tau in Bayesian approach can be used for:
  - flexible credible intervals
  - calculating exceedance probabilities of threshold values
  - comparative risk assessment
  - prediction of theoretical future studies (simulation)

Conclusions

- [ ] Conclusion, with reference to aim of study
- [ ] Implications of results: policy, (scientific?) practice, future research needs

List of Abbreviations

Abbreviation	Meaning
BLL	Blood lead levels
CI	Confidence interval
CrI	Credible interval
EBD	Environmental Burden of Disease
EU	European Union
IQ	Intelligence quotient
LLM	Large language model
REML	Restricted maximum-likelihood
RoB	Risk of Bias
SD	Standard deviation
SE	Standard error
UI	Uncertainty Interval
DALY	Disability-adjusted Life Year

Acknowledgements

The authors thank Rafiqa Benchrih (Department of Epidemiology and Public Health, Sciensano, Belgium), Jurgen Buekers (Flemish Institute for Technological Research, VITO, Belgium), Carla Martins (National School of Public Health, NOVA University of Lisbon, Portugal), Emily Mc Vey (Institute for Public Health and the Environment, RIVM, The Netherlands), Anthony Purece (Flemish Institute for Technological Research, VITO, Belgium) and Sofie Thomsen (Technical University of Denmark, DTU, Denmark) for their help in the screening process.

Funding

This work was carried out in the framework of the European Partnership for the Assessment of Risks from Chemicals (PARC) and has received funding from the European Union's Horizon Europe research and innovation programme under Grant Agreement No 101057014. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the Health and Digital Executive Agency. Neither the European Union nor the granting authority can be held responsible for them.

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